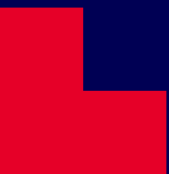

Artificial Intelligence

Week 8 (s. 7):
Reasoning under Uncertainty

COSC2129/2724/3118

A red geometric shape consisting of two overlapping squares, one slightly offset from the other, located in the bottom-left corner of the slide.

Outline

- Recap: Automated planning
- Uncertainty and rational decision
- Probabilities
 - Basic concepts
 - Unconditional and conditional probabilities
 - Probability distributions
 - Inference by enumeration
- Q&A

Warm up

- What's new in AI in the last few weeks?
- Mentimeter quizz

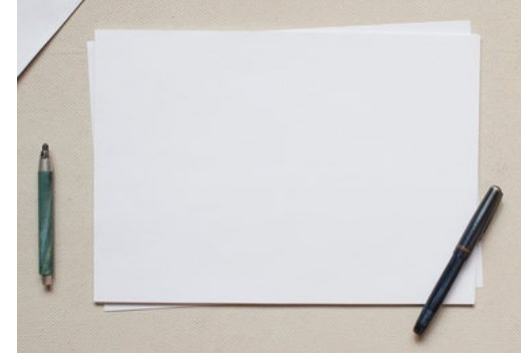
Prepare for Assessment 2: 5-6/9 (Week 10)

- Upcoming holidays: 30/8-02/9
- Assessment 2:
 - Undergrad: 5/9/25, Friday
 - MAI: 6/9/25, Saturday
- Need a make-up class:
 - Monday 25/8: Session 9
 - Wednesday 27/8: Session 10, as normal
- Tut/Lab:
 - Session 9: Fri. morning 29/8, as normal
 - Session 10: Wed. 03/9, as normal
 - MAI: Session 9?
 - Session 9: Fri. morning 29/8, offline, B2.4.02
 - or Fri. 29/8, hybrid

17	18 Monday		20 Wednesday	21	22 Fri	23 Sat
24	25 Session 9	26	27 Session 10	28 Tut MAI?	29 Tut/Lab 9	30 X
31	Sep 1	2	3 All students	4 Add. class	5 Test	6 MAI Test
			Tut/Lab 10			

In-class test

- Cover topics from week 4 to week 10
- Open book, **no AI tools**
- **Exam paper (with printed questions) will be provided**
- Bring your pen and paper for draft
- Write your answers clearly in the **provided exam paper**
 - **Do not** submit your draft or other papers (marking!)
- Submit your exam paper to the teaching staff in due time (no late submission is allowed).
- Questions?



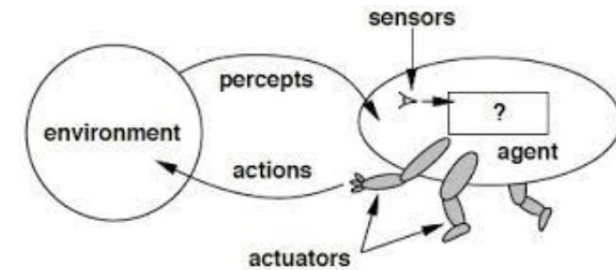
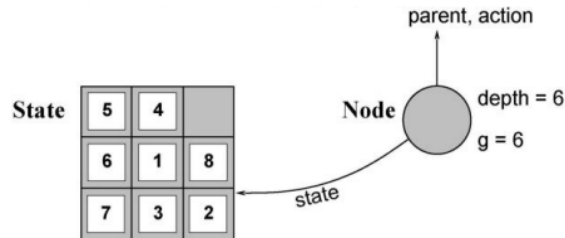
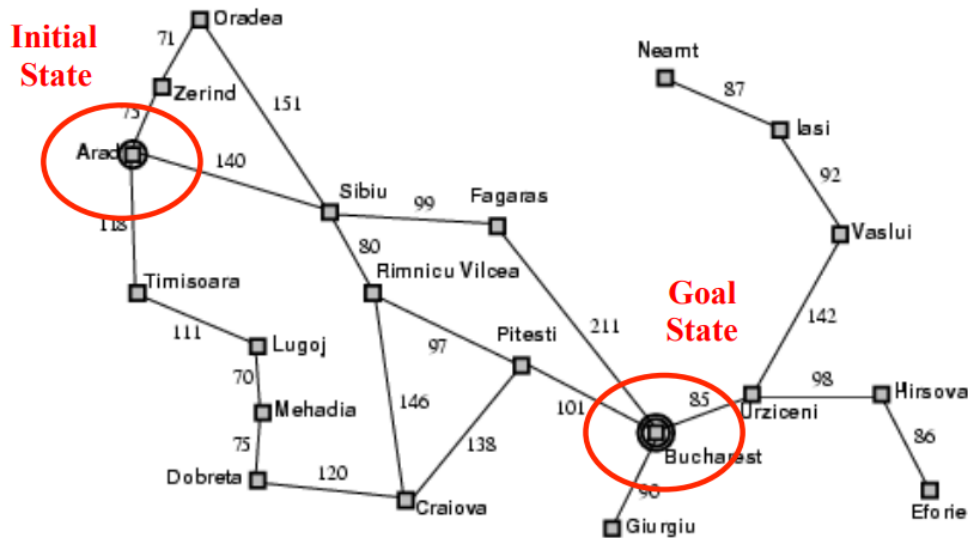
Automated planning

Materials:

- Book: [Artificial Intelligence: A Modern Approach](#)
 - Chapters 12-13 [reading: chapter 14]
- Other sources.

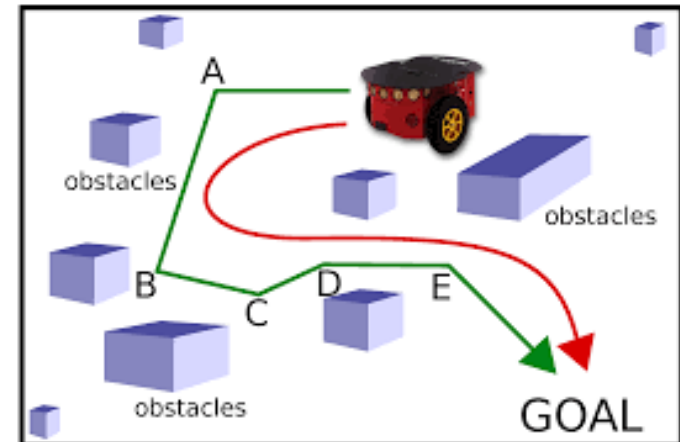
Search and planning

- Path finding



Automated Planning

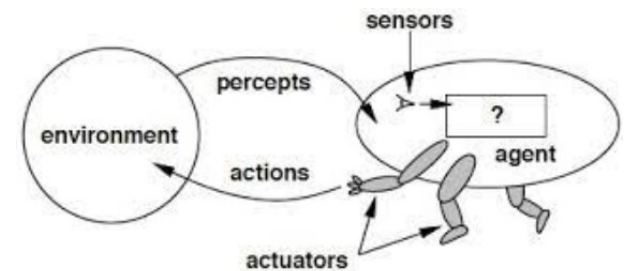
- What if there is traffic jam?
- What if the car broken?
- Accident, bridge closed, run out gas?
- ...



Uncertainty

- Agents need to handle uncertainty, whether due to **partial observability**, **non-determinism**, or a **combination** of the two.

- What if there is traffic jam?
- What if the car broken?
- Accident, bridge closed, run out gas?
- ...



Making decision under uncertainty

- Decision making scenarios
 - Considering to buy a car or a laptop
 - Taking a taxi to the airport



with: How to Buy a Car



$P(A_{25} \text{ gets me there on time} \mid \dots)$	$= 0.04$
$P(A_{90} \text{ gets me there on time} \mid \dots)$	$= 0.97$
$P(A_{120} \text{ gets me there on time} \mid \dots)$	$= 0.99$
$P(A_{1440} \text{ gets me there on time} \mid \dots)$	$= 0.999999$

Making decision under uncertainty

- Example: in PL/ FOL
 - Study hard $\Rightarrow$ passed exam
- Is it true?
 - High body temperature $\Rightarrow$ Fever
 - Fever $\Rightarrow$ Infection

Making decision under uncertainty

Diagnosis almost always involves uncertainty.

Logical reasoning breaks down when dealing with uncertainty.

A simple example – diagnosing a dental patient's toothache

-- A possible rule for dental diagnosis using first-order logic

$$\forall p \text{ Symptom}(p, \text{Toothache}) \Rightarrow \text{Disease}(p, \text{Cavity})$$

-- The above rule is wrong since not all patients with toothaches have cavities, so we change it to:

$$\forall p \text{ Symptom}(p, \text{Toothache}) \Rightarrow \text{Disease}(p, \text{Cavity}) \vee \text{Disease}(p, \text{GumDisease}) \\ \vee \text{Disease}(p, \text{ImpactedWisdom}) \vee \dots$$

-- The problem is that we could have an unlimited list. How about a causal rule:

$$\forall p \text{ Disease}(p, \text{Cavity}) \Rightarrow \text{Symptom}(p, \text{Toothache})$$

-- The above rule is still incorrect, since not all cavities cause pain.

Solution: making it **logically exhaustive** – augmenting the left-hand side with all the qualifications required.

Making decision under uncertainty

- Uncertainty: Partial observability, nondeterminism, or both.
- Solution: keeping track of **all possible world states**, and generating **a contingent plan** that handles every possible eventuality.
- The right thing to do, i.e., **the rational decision**, depends on both the relative importance of various goals and the likelihood that, and degree to which, they will be achieved.

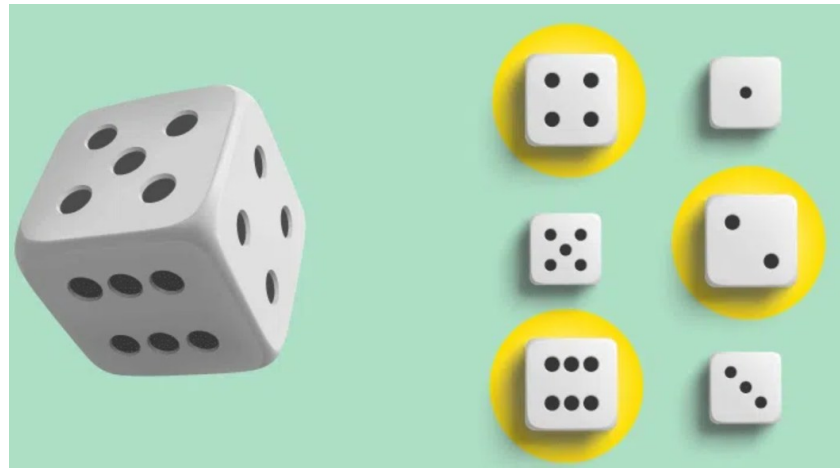
Decision Theory

Decision Theory = Probability Theory + Utility Theory

- Fundamental idea: an agent is rational if and only if it chooses the action that yields the **highest expected utility**, averaged over all the possible outcomes of the action. -- Principle of **m**aximum **e**xpected **u**tility (MEU)

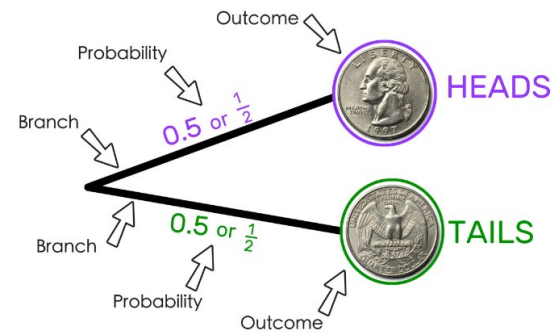
Probability Basics

- Random variables
- Syntax



Basic Probability

- Tossing a coin



- Sample space: $\Omega = \{\text{Head}, \text{Tail}\}$
- Event: Head, tail

Basic Probability

- Rolling a die



ROLL					
					
1	2	3	4	5	6
TOTAL					

- Sample space: Ω ?
- Events: subset of Ω
 - The outcome is 3
 - The outcome is <3
 - Two consecutive outcomes are equal

Basic Probability

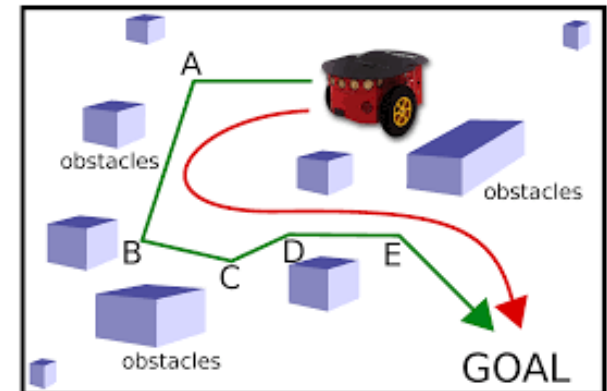
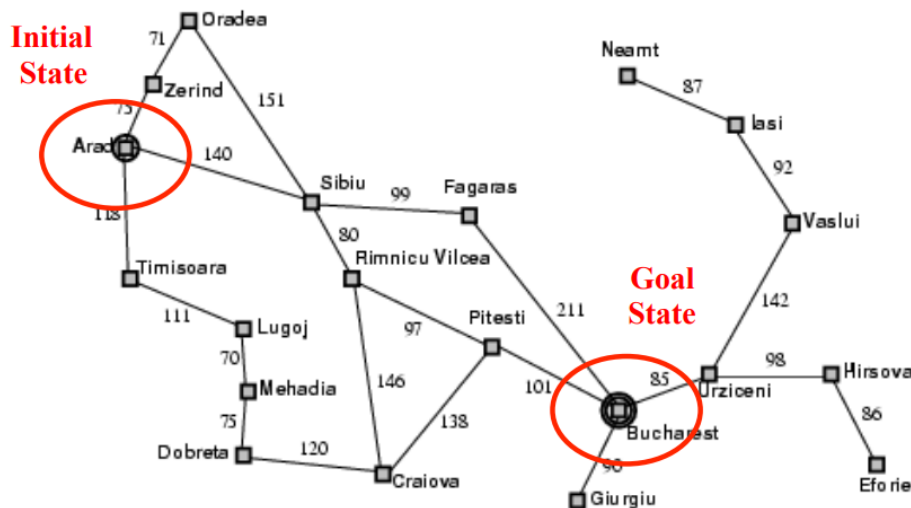
- Rolling two dice

										
					 					
										
										
										
2	3	4	5	6	7	8	9	10	11	12

- Sample space: Ω ?
- Events: subset of Ω
 - The outcome is 3
 - The outcome is < 3
 - Two consecutive outcomes are equal

Example: Student being on time

- The probability of arriving on time when taking a taxi is 0.6.
- The probability of arriving on time when taking a bus 0.2
- A student would take a taxi more frequent than taking a bus. On average, for every 10 days, 7 of them are with taxi (0.7 probability) and 3 of them are with bus (0.3 probability).
- What is the probability that he will be ontime in a given day?

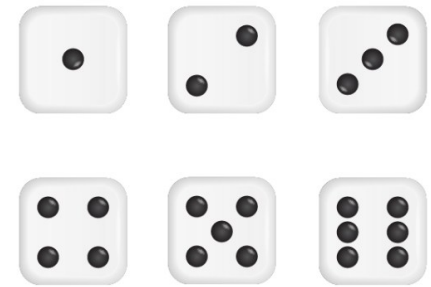


Basic probability

- **Sample space** (the set of all possible worlds): Ω
e.g., 6 possible worlds of rolling a die.
 $\omega \in \Omega$ is any **possible world**, e.g., rolled number is 1.
The possible worlds are *mutually exclusive* and *exhaustive*
- **Event**: a set of possible worlds

$$P(A) = \sum_{\omega \in A} P(\omega)$$

E.g., $P(\text{DieRoll} < 4) = ?$



= {1, 2, 3, 4, 5, 6}

Variables

- Random variables with their own domains
 - Boolean random variables
E.g., *Cavity* (do I have a cavity?)
 - Discrete random variable (finite or infinite)
E.g., *Weather* is one of (*sunny*, *rain*, *cloudy*, *snow*)
Weather = *rain* is a proposition
*Values must mutually exclusive and be exhaustive
 - Continuous random variables (bounded or unbounded)
E.g., *Temperature* = 21.6
Temperature < 22.0 is also allowed
 - Arbitrary combination of basic propositions via logical connectives
E.g., (*Temperature* < 22.0) $\wedge$ (*Weather* = *rain*)

Probability Basics

- **Prior Probability:**
 - Prior or unconditional probabilities of propositions e.g., $P(\text{Cavity} = \text{true}) = 0.1$ and $P(\text{Weather} = \text{sunny}) = 0.72$
 - They refer to degrees of belief in propositions in the absence of any other information
- **Probability distribution** of a random variable gives the probabilities for all the possible values of this random variable.
 - $P(\text{Weather}) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$ (normalized, i.e., sums to 1)
- **Joint probability distribution** on a set of random variables gives the probabilities of all combinations of the values of those variables.
 - $P(\text{Weather}, \text{Cavity})$ is a 4×2 table of probabilities

<i>Weather =</i>	<i>sunny</i>	<i>rainy</i>	<i>cloudy</i>	<i>snow</i>
<i>Cavity = true</i>	0.144	0.02	0.016	0.02
<i>Cavity = false</i>	0.576	0.08	0.064	0.08

Conditional Probability

- Definition: Conditional or *posterior* probabilities
$$P(a \mid b) = P(a \wedge b) / P(b) \text{ if } P(b) > 0$$
 - e.g., $P(\text{cavity} \mid \text{toothache}) = 0.8$, i.e., given that toothache is known
- Condition distributions: $P(X \mid Y)$ gives the values of $P(X = x_i \mid Y = y_j)$ for each possible (i, j) pair.
 - E.g., $P(\text{Cavity} \mid \text{Toothache})$ is 2-element vector of 2-element vectors

Conditional Probability

- **Product rule** gives an alternative formulation
 - $P(a \wedge b) = P(a | b) P(b) = P(b | a) P(a)$
- Chain rule is derived by successive application of product rule:

$$\begin{aligned} P(X_1, \dots, X_n) &= P(X_1, \dots, X_{n-1}) P(X_n | X_1, \dots, X_{n-1}) \\ &= P(X_1, \dots, X_{n-2}) P(X_{n-1} | X_1, \dots, X_{n-2}) P(X_n | X_1, \dots, X_{n-1}) \\ &= \dots \\ &= P(X_1) \prod_{i=2}^n P(X_i | X_1, \dots, X_{i-1}) \end{aligned}$$

Inference by Enumeration

- Probabilistic inference: Computing posterior probabilities for *query* propositions given observed evidences

Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

For any proposition ϕ , sum over all possible worlds where it holds:

$$P(\phi) = \sum_{\omega \in M(\phi)} P(\omega)$$

Inference by Enumeration

- Example: Do you remember this question
 - rolling a die and weather: $p(\text{outcome} = 3, \text{sunny}) = ?$

Inference by Enumeration: General case

- General case:
 - Evidence variables: $E_1, \dots, E_k = e_1, \dots, e_k$
 - Query variable: Q
 - Hidden variables: $H_1, \dots, H_r$
 - $\{E_1, \dots, E_k\} + Q + \{H_1, \dots, H_r\} \longrightarrow$ all variables
- Query:
 - $P(Q|e_1, \dots, e_k)$, works fine with multiple query variables,
 - **Capital** letter represents a variable, X means the variable X
 - *Small* letter represents the variable takes certain value/state, x means $X=x$
 - X may have multiple possible values: e.g X is a weather variable, then (sunny, rainy, cloudy, snowy)

Practice

- Example 1

- Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- For any proposition ϕ , sum over all possible worlds where it holds:

$$P(\phi) = \sum_{\omega \in M(\phi)} P(\omega)$$

$$P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$

Practice

- Example 1

- Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

- Can also compute conditional probabilities:

$$P(\neg \text{cavity} \mid \text{toothache})$$

$$= \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$

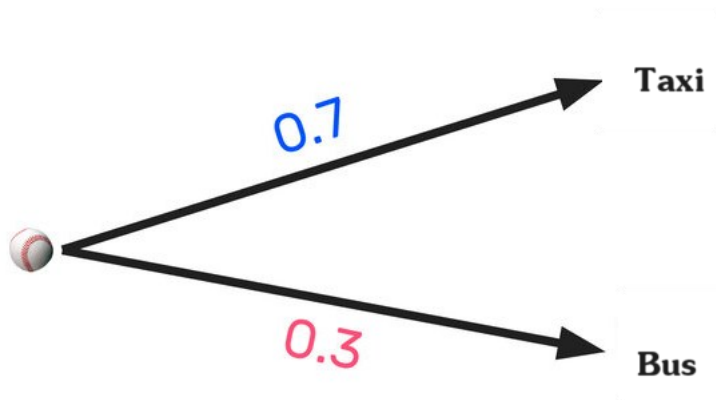
$$= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4$$

Example 2: Student being on time

- The probability of arriving on time when taking a taxi is 0.6.
- The probability of arriving on time when taking a bus is 0.2.
- A student would take a taxi more frequent than taking a bus. On average, for every 10 days, 7 of them are with taxi (0.7 probability) and 3 of them are with bus (0.3 probability).
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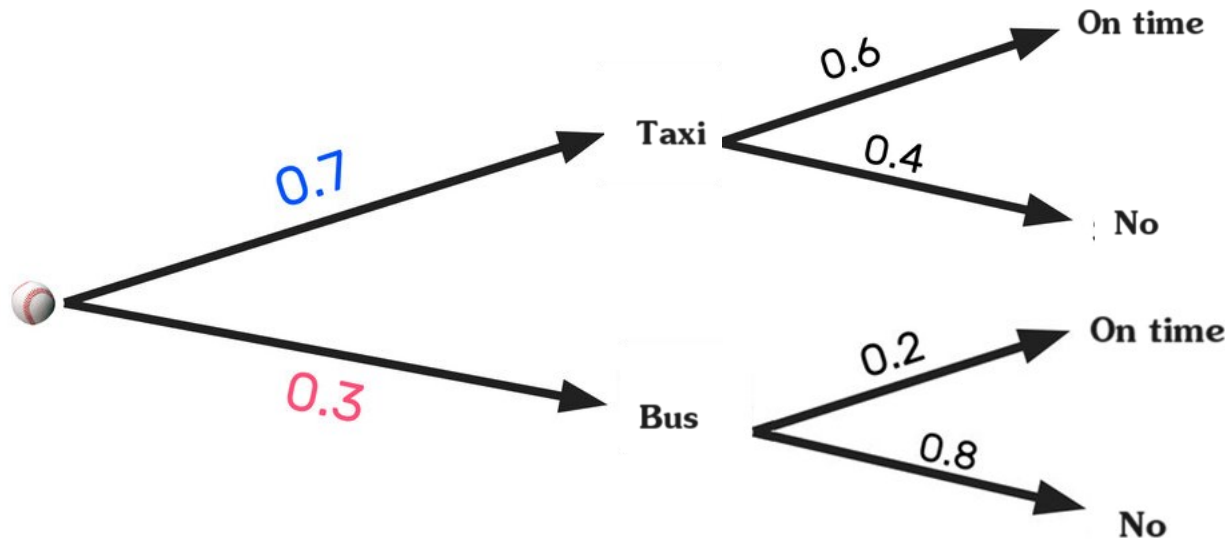
Example: Student being on time

- On time – taxi: 0.6.
- On time – bus: 0.2
- Prob taxi: 0.7.
- Prob bus: 0.3
- What is the probability that he will be ontime in a given day?



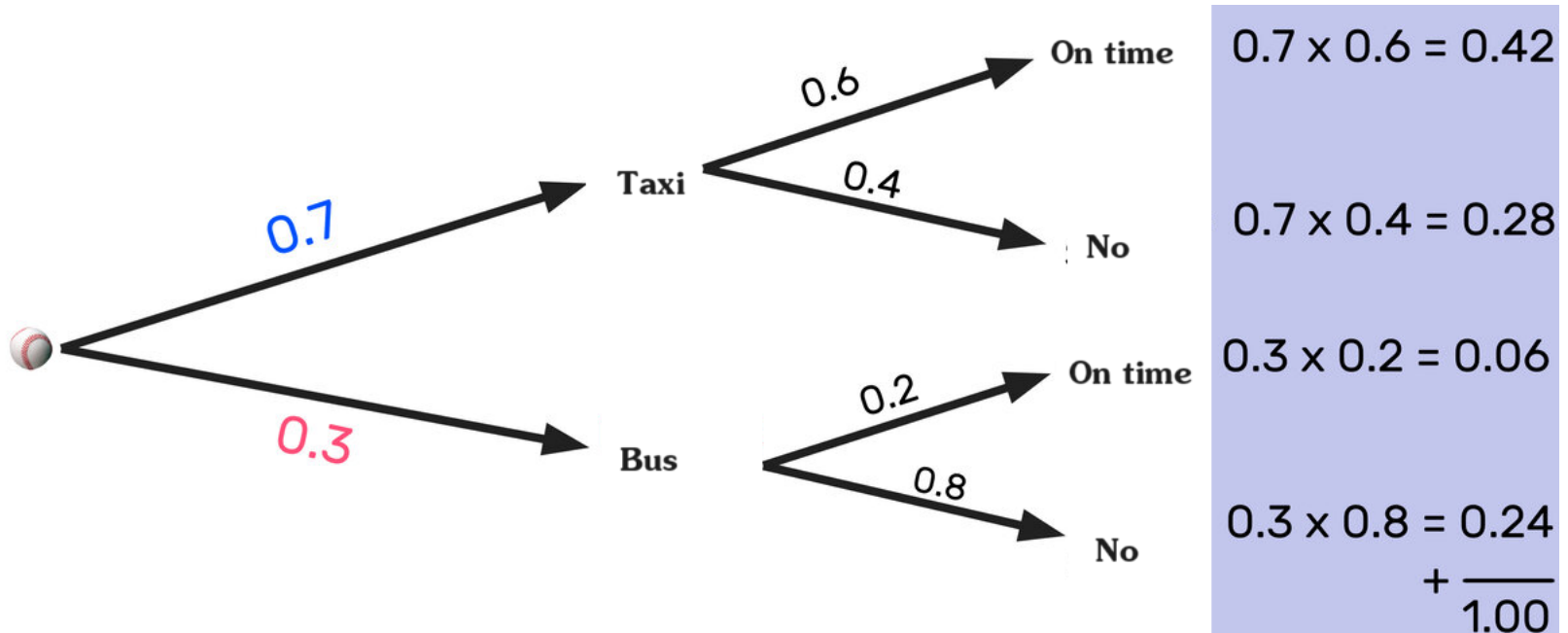
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Example: Student being on time

- On time – taxi: 0.6.
- On time – bus: 0.2
- Prob taxi: 0.7.
- Prob bus: 0.3
- What is the probability that he will be ontime in a given day?



Example 2

- Scenario: You receive two job offers. One is from a stable company with a high salary, but you're not sure if you'll enjoy the work. The other is from a startup with lower pay but more exciting and creative work, though there's a higher risk of failure and job insecurity.
- Job A (Stable, High Salary):
 - 90% chance of job stability and a high salary for the next 5 years.
 - 10% chance of a downturn, with possible job loss.
- Job B (Startup, Lower Pay, Risky):
 - 70% chance the startup will succeed, providing good career growth and fulfillment.
 - 30% chance the startup fails, and you lose your job.
- You assign a utility value to each possible outcome based on your preferences. Based on expected utility, which job is the better choice?

Example 2: solution

- Assign utilities
 - For job A:
 - If stable, you value it at 8/10 (high salary, but not as fulfilling).
 - If unstable (job loss), you value it at 3/10 (because of uncertainty).
 - For job B:
 - If successful, you value it at 9/10 (because of fulfillment and potential growth).
 - If it fails, you value it at 2/10 (due to job loss and financial instability).
- Calculate Expected Utility

Job A:

$$\text{Expected Utility} = (0.9 \times 8) + (0.1 \times 3) = 7.2 + 0.3 = 7.5$$

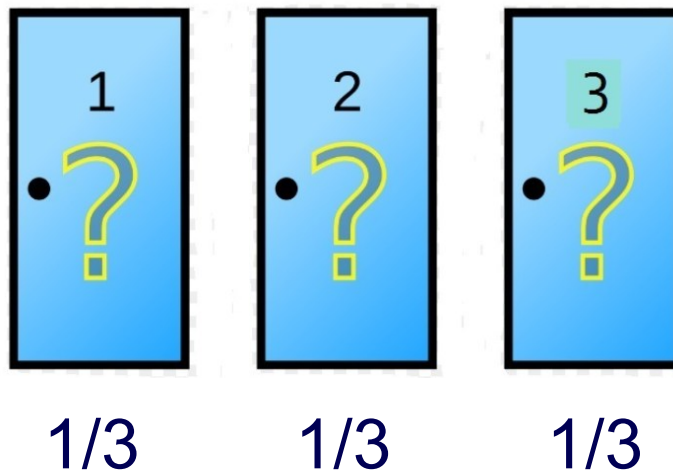
Job B:

$$\text{Expected Utility} = (0.7 \times 9) + (0.3 \times 2) = 6.3 + 0.6 = 6.9$$

- Note: the result can be different! Why?

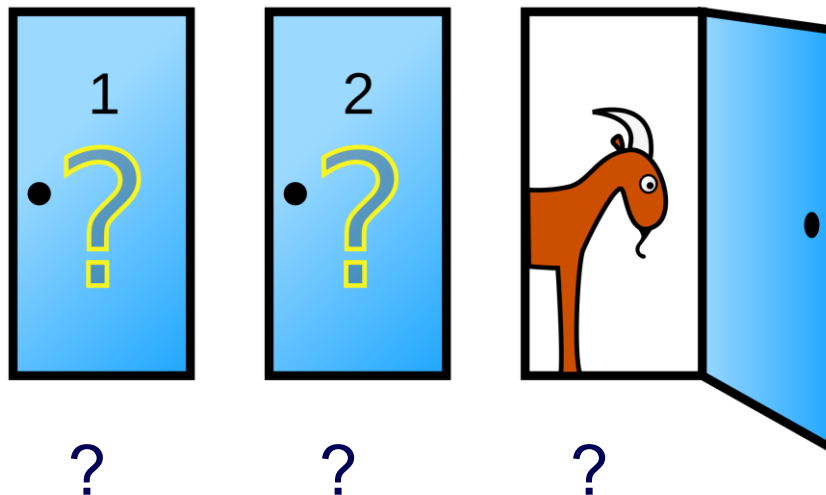
Example 3: The Monty Hall problem - a probability puzzle

- Suppose you're on a game show, and you're given the choice of 3 doors:
 - Behind one door is **a car**; behind the others, **goats**.
- You pick a door, say **No. 1**, and the host, who knows what's behind the doors, opens another door, say **No. 3**, which has **a goat**.
- He then says to you, "Do you want to pick door No. 2?"
- Is it to your advantage to switch your choice?



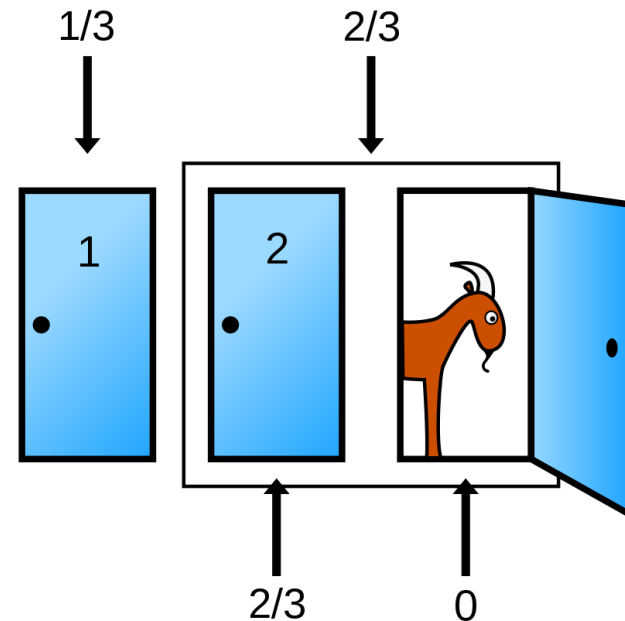
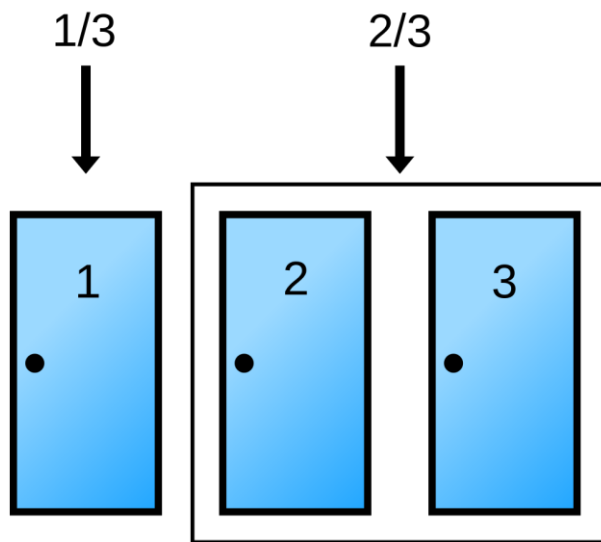
The Monty Hall problem - a probability puzzle

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- He then says to you, "Do you want to pick door No. 2?"
- Is it to your advantage to switch your choice?



The Monty Hall problem - a probability puzzle

- Do you want to change your mind?



The Monty Hall problem - a probability puzzle



Switch
and win



Switch
and win



Stay
and win



Player choice
before door is open

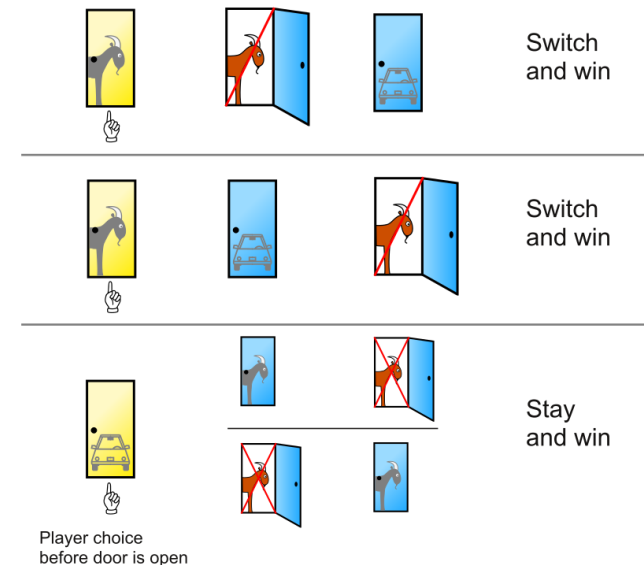
The Monty Hall problem - a probability puzzle

- Simple solution

Behind door 1	Behind door 2	Behind door 3	Result if staying at door #1	Result if switching to the door offered
Goat	Goat	Car	Wins goat	Wins car
Goat	Car	Goat	Wins goat	Wins car
Car	Goat	Goat	Wins car	Wins goat

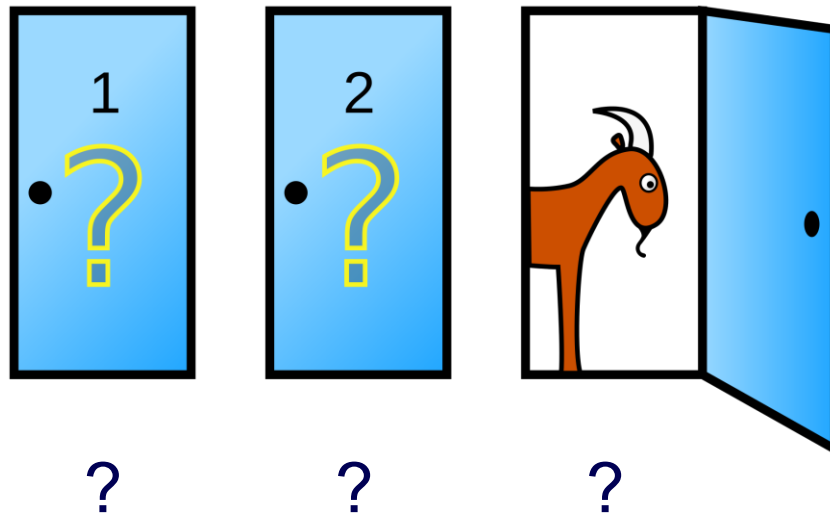
- Standard assumptions

- The host must always open a door that was not picked by the contestant.
- The host must always open a door to reveal a goat and never the car.
- The host must always offer the chance to switch between the originally chosen door and the remaining closed door.



The Monty Hall problem - a probability puzzle

- Will you switch your choice?



Example 4: Passing the final exam

- Enumerate possible outcomes:
 - Study time: Low, Moderate, High (3 possibilities)
 - Sleep: Inadequate, Adequate, Optimal (3 possibilities)
 - Stress level: Low, Moderate, High (3 possibilities)
 - There are ? possible combinations of these factors.

Example: Passing the final exam

- Assign probabilities to each combination based on real-world observations or data
- ...
- Define Success (Passing the Exam)
- ...
- Calculate Probability of Passing the Final Exam: Sum up the probabilities of passing the final exam.

Hand-on Practice

- [AI22 Tut07](#) (tut/lab session)

Question?